

Machine Test 05 – Process Management

Objectives

1. Display the currently logged-in user.

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ whoami  
sreelekha  
sreelekha@LAPTOP-CKI0PC9R:~$
```

2. Display the processes running in the current terminal session.

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ ps  
  PID TTY          TIME CMD  
 1606 pts/0    00:00:00 bash  
 1975 pts/0    00:00:00 ps  
sreelekha@LAPTOP-CKI0PC9R:~$
```

3. Display all running processes with detailed information.

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ ps aux  
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND  
root         1  0.0  0.4 24564 15556 ?        Ss   09:11   0:01 /sbin/init  
root         8  0.0  0.0 3180 2840 hvc0   S1+  09:11   0:00 plan0 --root  
0.3 22416 14496 ?        Ss   09:11   0:00 /usr/libroot  
Ss 09:11  0:00 /bin/sh  
root       110  0.0  0.0 4472 3184 ?        Ss   09:11   0:00 /usr/sbimessage+  
0.7 44100 29808 ?        Ss   09:11   0:00 /usr/binroot  
Ss1 09:11  0:00 /usr/sbiroot  
/usr/sbi_chrony 247  0.0  0.0 12096 2484 ?        S   09:11   0:00 /usr/sbiroot  
S10 0.0  0.1 8648 5024 ?        Ss   09:11   0:00 login --sreelek+  
4184 ?        S   09:11   0:00 (sd-pam)sreelek+  
root      1604  0.0  0.0 3184 1120 ?        Ss  10:16   0:00 /init  
root      1605  0.0  0.0 3200 1256 ?        S   10:16   0:00 /init  
sreelek+  1606  0.0  0.1 6268 5672 pts/0   Ss  10:16   0:00 -bash  
sreelek+  1998  0.0  0.1 7160 4420 pts/0   R+  10:55   0:00 ps aux  
sreelekha@LAPTOP-CKI0PC9R:~$
```

4. Find all Bash shell processes running on the system.

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ ps aux |grep bash  
sreelek+  391  0.0  0.1 6136 5612 pts/1   Ss+  09:11   0:00 -bash  
sreelek+  1606  0.0  0.1 6268 5672 pts/0   Ss   10:16   0:00 -bash  
sreelek+  2021  0.0  0.0 4128 2476 pts/0    R+   10:56   0:00 grep --color=auto bash  
sreelekha@LAPTOP-CKI0PC9R:~$
```

5. Install the htop package.

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ sudo apt install htop  
[sudo: authenticate] Password:  
Installing:  
  htop  
Suggested packages:  
  lm-sensors strace  
Summary:  
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 11  
  Download size: 177 kB  
  Space needed: 451 kB / 1024 GB available  
Get:1 http://archive.ubuntu.com/ubuntu/resolute/main amd64 htop amd64 3.4.1-5build2 [177 kB]  
Fetched 177 kB in 1s (140 kB/s)  
Selecting previously unselected package htop.  
(Reading database ... 41926 files and directories currently installed.)  
Preparing to unpack .../htop-3.4.1-5build2_amd64.deb ...  
Unpacking htop (3.4.1-5build2) ...  
Setting up htop (3.4.1-5build2) ...  
Processing triggers for hicolor-icon-theme (0.18-2build1) ...  
Processing triggers for man-db (2.13.1-1build1) ...  
sreelekha@LAPTOP-CKI0PC9R:~$
```

6. Launch htop and verify that CPU and Memory usage are visible.

```
0.0% 4[ 0.0%]
1[ 0.0%] 5[ 0.0%]
2[ 0.0%] 6[ 0.0%]
3[ 0.0%] 7[ 0.7%]
Mem[|||||] 385M/3.65G Tasks: 26, 12 thr, 0 kthr; 1 running
Sup[ 0K/1.00G Load average: 0.04 0.03 0.00
Uptime: 01:47:24
```

PID	USER	PRI	NI	VIRT	RES	SHR	S	CPU%	MEM%	TIME+	Command
1	root	20	0	24564	15556	11636	S	0.0	0.4	0:01.15	/sbin/init
2	root	20	0	3188	2288	2072	S	0.0	0.1	0:00.01	/init
8	root	20	0	3188	2040	1940	S	0.0	0.1	0:00.00	plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
9	root	20	0	3188	2040	0	S	0.0	0.1	0:00.00	plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
10	root	20	0	3188	2288	0	S	0.0	0.1	0:00.00	/init
48	root	19	-1	50404	17016	15736	S	0.0	0.4	0:00.37	/usr/lib/systemd/systemd-journald
80	systemd-re	20	0	22416	14496	12000	S	0.0	0.4	0:00.15	/usr/lib/systemd/systemd-resolved
91	root	20	0	35276	12312	9320	S	0.0	0.3	0:01.18	/usr/lib/systemd/systemd-udevd
109	root	20	0	2888	2076	1940	S	0.0	0.1	0:00.05	/bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
110	root	20	0	4472	3184	2940	S	0.0	0.1	0:00.02	/usr/sbin/cron -f -P
111	messagebus	20	0	8820	5584	4900	S	0.0	0.1	0:00.10	@dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
121	root	20	0	44100	29888	14672	S	0.0	0.8	0:00.29	/usr/bin/networkd-dispatcher --run-startup-triggers
139	root	20	0	18444	9456	8200	S	0.0	0.2	0:00.12	/usr/sbin/systemd-logind
174	syslog	20	0	215M	5924	4628	S	0.0	0.2	0:00.07	/usr/sbin/rsyslogd -n -iNONE
189	root	20	0	5492	2852	2668	S	0.0	0.1	0:00.01	/usr/sbin/agetty --noneset --noclear --issue-file=/etc/issue:/etc/issue.d:/run/issue.d:/usr/lib/issue.d
204	syslog	20	0	215M	5924	0	S	0.0	0.2	0:00.02	/usr/sbin/rsyslogd -n -iNONE
205	syslog	20	0	215M	5924	0	S	0.0	0.2	0:00.00	/usr/sbin/rsyslogd -n -iNONE
209	syslog	20	0	215M	5924	0	S	0.0	0.2	0:00.01	/usr/sbin/rsyslogd -n -iNONE
233	_chrony	20	0	23588	11232	7284	S	0.0	0.3	0:00.34	/usr/sbin/chronyd -n -F 1 -x
247	_chrony	20	0	12096	2484	1760	S	0.0	0.1	0:00.00	/usr/sbin/chronyd -n -F 1 -x
272	root	20	0	120M	32712	18100	S	0.0	0.9	0:00.18	/usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
279	root	20	0	120M	32712	0	S	0.0	0.9	0:00.00	/usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
319	root	20	0	8648	5624	4780	S	0.0	0.1	0:00.00	login -- sreelekha
361	sreelekha	20	0	22764	13396	10932	S	0.0	0.4	0:00.08	/usr/lib/systemd/systemd --user
363	sreelekha	20	0	22916	4184	2104	S	0.0	0.1	0:00.00	(sd-pam)
391	sreelekha	20	0	6136	5612	4056	S	0.0	0.1	0:00.03	-bash
1604	root	20	0	3184	1120	980	S	0.0	0.0	0:00.00	/init
1605	root	20	0	3200	1236	1108	S	0.0	0.0	0:00.21	/init

7. Open a new terminal and create a process that runs for 5 minutes. 300 seconds = 5 minutes.

```
sreelekha@LAPTOP-CKI0PC9R:~$ sleep 300
```

8. Search for the process created in Step 7.

```
sreelekha@LAPTOP-CKI0PC9R:~$ ps aux | grep "sleep 300"
sreelek+ 2517 0.0 0.2 16112 7860 pts/0 S+ 11:27 0:00 sleep 300
sreelek+ 2524 0.0 0.0 4260 2476 pts/2 S+ 11:28 0:00 grep --color=auto sleep 300
sreelekha@LAPTOP-CKI0PC9R:~$
```

9. Identify the PID of the process.

PID is 2517

10. Open the real-time process monitoring tool and observe running processes.

```

sreelekha@LAPTOP-CKI0PC9R: ~
0% 0.0% 4[ 0.0%]
1[ 0.0%] 5[ 0.0%]
2[ 0.0%] 6[ 0.0%]
3[ 0.7%] 7[ 0.0%]
Mem[|||||]
Swp[|||||]
388M/3.65G Tasks: 29, 9 thr, 0 kthr; 1 running
0K/1.00G Load average: 0.00 0.02 0.00
Uptime: 02:17:15

Main 1/0
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
1 root 20 0 24564 15556 11636 S 0.0 0.4 0:01.21 /sbin/init
2 root 20 0 3188 2288 2872 S 0.0 0.1 0:00.01 /init
8 root 20 0 3188 2040 1940 S 0.0 0.1 0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
9 root 20 0 3188 2040 0 S 0.0 0.1 0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
10 root 20 0 3188 2288 0 S 0.0 0.1 0:00.00 /init
48 root 19 -1 58404 17060 15748 S 0.0 0.4 0:00.43 /usr/lib/systemd/systemd-journald
80 systemd-re 20 0 22416 14496 12000 S 0.0 0.4 0:00.15 /usr/lib/systemd/systemd-resolved
91 root 20 0 35276 12312 9320 S 0.0 0.3 0:01.60 /usr/lib/systemd/systemd-udevd
109 root 20 0 2888 2076 1948 S 0.0 0.1 0:00.05 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -f 1
110 root 20 0 4472 2184 2940 S 0.0 0.1 0:00.02 /usr/sbin/cron -f -p
111 messagebus 20 0 8820 5584 4900 S 0.0 0.1 0:00.11 dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
121 root 20 0 44100 25808 14672 S 0.0 0.8 0:00.29 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
139 root 20 0 18444 9456 8208 S 0.0 0.2 0:00.13 /usr/lib/systemd/systemd-logind
174 syslog 20 0 215M 5924 4628 S 0.0 0.2 0:00.07 /usr/sbin/rsyslogd -n -iNONE
189 root 20 0 5492 2852 2668 S 0.0 0.1 0:00.01 /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:/etc/issue.d/run/issue.d:/usr/lib/issue.d
204 syslog 20 0 215M 5924 0 S 0.0 0.2 0:00.03 /usr/sbin/rsyslogd -n -iNONE
205 syslog 20 0 215M 5924 0 S 0.0 0.2 0:00.00 /usr/sbin/rsyslogd -n -iNONE
209 syslog 20 0 215M 5924 0 S 0.0 0.2 0:00.01 /usr/sbin/rsyslogd -n -iNONE
233 _chrony 20 0 23588 11232 7284 S 0.0 0.3 0:00.36 /usr/sbin/rsyslogd -n -iNONE
247 _chrony 20 0 12096 2484 1760 S 0.0 0.1 0:00.00 /usr/sbin/chronyd -n -f 1 -x
272 root 20 0 120M 32712 18100 S 0.0 0.9 0:00.18 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
279 root 20 0 120M 32712 0 S 0.0 0.9 0:00.00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
319 root 20 0 8648 5624 4780 S 0.0 0.1 0:00.00 login -- sreelekha
361 sreelekha 20 0 22764 13396 10932 S 0.0 0.4 0:00.08 /usr/lib/systemd/systemd --user
363 sreelekha 20 0 22916 4104 2104 S 0.0 0.1 0:00.00 (sd-pam)
391 sreelekha 20 0 6136 5612 4856 S 0.0 0.1 0:00.03 -bash
2132 polkitd 20 0 299M 8520 7784 S 0.0 0.2 0:00.12 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
2134 polkitd 20 0 299M 8520 0 S 0.0 0.2 0:00.06 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice

```

Press F10 or q to exit.

11. Terminate the process created in Step 7 using its PID.

```

sreelekha@LAPTOP-CKI0PC9R: ~
sreelekha@LAPTOP-CKI0PC9R:~$ ps aux |grep "sleep 300"
sreelek+ 2517 0.0 0.2 16112 7860 pts/0 S+ 11:27 0:00 sleep 300
sreelek+ 2524 0.0 0.0 4260 2476 pts/2 S+ 11:28 0:00 grep --color=auto sleep 300
sreelekha@LAPTOP-CKI0PC9R:~$ http
sreelekha@LAPTOP-CKI0PC9R:~$ kill 2517
sreelekha@LAPTOP-CKI0PC9R:~$

```

12. Verify that the process has been terminated successfully.

```

sreelekha@LAPTOP-CKI0PC9R: ~
sreelekha@LAPTOP-CKI0PC9R:~$ sleep 300
Terminated sleep 300
sreelekha@LAPTOP-CKI0PC9R:~$

```

13. Create another process that runs for 10 minutes.

```

sreelekha@LAPTOP-CKI0PC9R: ~
sreelekha@LAPTOP-CKI0PC9R:~$ sleep 600

```

14. Find the PID of the new process.

PID:2575

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ ps aux |grep "sleep 600"  
sreelek+ 2575 0.0 0.2 16112 7980 pts/0 S+ 11:31 0:00 sleep 600  
sreelek+ 2592 0.0 0.0 4260 2508 pts/2 S+ 11:32 0:00 grep --color=auto sleep 600  
sreelekha@LAPTOP-CKI0PC9R:~$
```

15. Forcefully terminate the process using the appropriate command.

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ ps aux |grep "sleep 600"  
sreelek+ 2575 0.0 0.2 16112 7980 pts/0 S+ 11:31 0:00 sleep 600  
sreelek+ 2592 0.0 0.0 4260 2508 pts/2 S+ 11:32 0:00 grep --color=auto sleep 600  
sreelekha@LAPTOP-CKI0PC9R:~$ kill -9 2575  
sreelekha@LAPTOP-CKI0PC9R:~$
```

16. Verify that the process is no longer running.

```
Select sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ sleep 600  
Killed sleep 600  
sreelekha@LAPTOP-CKI0PC9R:~$ ps -p 2575  
PID TTY TIME CMD  
sreelekha@LAPTOP-CKI0PC9R:~$
```

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ ps aux |grep "sleep 600"  
sreelek+ 2575 0.0 0.2 16112 7980 pts/0 S+ 11:31 0:00 sleep 600  
sreelek+ 2592 0.0 0.0 4260 2508 pts/2 S+ 11:32 0:00 grep --color=auto sleep 600  
sreelekha@LAPTOP-CKI0PC9R:~$ kill -9 2575  
sreelekha@LAPTOP-CKI0PC9R:~$ ps -p 2575  
PID TTY TIME CMD  
sreelekha@LAPTOP-CKI0PC9R:~$
```